Lecture 8. Estimation of Basic Quantities for Other Sampling Schemes (Part 2)

Double Censoring, Interval Censoring, Cohort Life Table

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1 Other Sampling Schemes

- Interval censoring: $X_i \in (L_i, R_i]$, never observed exactly.
- Double censoring: each observation is LC, RC, or exact.
- LT + interval censoring, or LT + double censoring.

Estimation here is hard: there is no simple closed-form estimator, and inference is not easy.

2 Turnbull's Iterative Method (1974) for Double Censoring

Pure left censoring is rare; samples with both left and right censoring are more common. Turnbull (1974) proposed a modified product-limit estimator with no closed form, computed iteratively.

- **Idea.** Start with an estimate of S that ignores the left-censored data; use it to compute the *expected* number of deaths at each observed time contributed by the left-censored observations; update S; repeat until it stabilizes (a self-consistency / EM-type algorithm).
- **Notation.** $0 = t_0 < t_1 < \cdots < t_m$ the distinct observed times; d_j = deaths at t_j (possibly 0); c_j = left-censored observations at t_j; r_j = right-censored observations at t_j.

Definition 8.1: Turnbull (1974) algorithm

Step 0. Initial estimate $S_0(t_j)$, e.g. the KM ignoring the left-censored observations.

Step $(K{+}1).1$**.** For each left-censoring time t_i and $j \le i$, estimate $p_{ij} = P(t_{j-1} < X \le t_j \mid$

$X \le t_i$) by

$$p_{ij} = \frac{S_K(t_{j-1}) - S_K(t_j)}{1 - S_K(t_i)}.$$

Step $(K+1).2$**.** Estimated events at t_j: $\hat{d}_j = d_j + \sum_{i=j}^{m} c_i\, p_{ij}$ (each left-censored observation is distributed over the times before it).

Step $(K+1).3$**.** Compute the usual KM from the pseudo right-censored data with $\hat{d}_j$ events and r_j right-censored observations at t_j, ignoring the left-censored data: $Y_j = \sum_{l=j}^{m}(\hat{d}_l + r_l)$.

Repeat until $S_{K+1}(t_j) \approx S_K(t_j)$ for all t_j.

Remark 8.2: Why p_{ij} has this form

A boy left censored at t_i is known to have $X \le t_i$. Given that, the chance his event fell in $(t_{j-1}, t_j]$ is $P(t_{j-1} < X \le t_j)/P(X \le t_i) = [S(t_{j-1}) - S(t_j)]/[1 - S(t_i)]$. Replacing each left-censored observation by these fractional events is the E-step; recomputing KM is the M-step.

Example 8.3: Age at first marijuana use (K&M Section 1.17)

California high-school boys: left censored = used marijuana but cannot recall the age of first use; right censored = never used. Initial KM ignoring the 12 left-censored boys (K&M Table 5.1):

t_i	10	11	12	13	14	15	16	17	18	> 18	Total
c_i	0	0	0	1	2	3	2	3	1	0	12
d_i	4	12	19	24	20	13	3	1	0	4	100
r_i	0	0	2	15	24	18	14	6	0	0	79
Y_i	179	175	163	142	103	59	28	11	4	4	
$S_0(t_i)$	.978	.911	.804	.669	.539	.420	.375	.341	.341	0	

Step 1 at $t_4 = 13$ (one left-censored boy, $1 - S_0(13) = 0.331$):

$$p_{41} = \tfrac{1.000-0.978}{0.331} = 0.067, \quad p_{42} = \tfrac{0.978-0.911}{0.331} = 0.202,$$

$$p_{43} = \tfrac{0.911-0.804}{0.331} = 0.320, \quad p_{44} = \tfrac{0.804-0.669}{0.331} = 0.410.$$

Doing this for every i with $c_i > 0$ (K&M Table 5.2; e.g. the column $i = 9$ reads $0.034, 0.102, 0.161, 0.206, 0.197, 0.180, 0.068, 0.052, 0.000$), Step 2 gives

$$\hat{d}_1 = 4 + 0.067(1) + 0.048(2) + 0.039(3) + 0.036(2) + 0.034(3) + 0.034(1) = 4.487.$$

Step 3 (K&M Table 5.3).

t_i	10	11	12	13	14	15	16	17	18	> 18
$\hat{d}_i$	4.487	13.461	21.313	26.963	22.437	14.714	3.417	1.207	0	4
Y_i	191.000	186.513	173.052	149.739	107.775	61.338	28.624	11.207	4	4
$S_1(t_i)$	.977	.906	.794	.651	.516	.392	.345	.308	.308	0

Iterate until convergence. Note $Y_1 = 191 = 179 + 12$: all left-censored boys are now accounted for as (fractional) events.

3 Turnbull's Iterative Method (1976) for Interval-Censored Data

Let $0 = \tau_0 < \tau_1 < \cdots < \tau_m$ be a grid containing all L_i, R_i. Define $\alpha_{ij} = 1$ if $(\tau_{j-1}, \tau_j] \subseteq (L_i, R_i]$ and 0 otherwise, i.e. whether the event of subject i could have occurred at τ_j.

Definition 8.4: Turnbull (1976) algorithm

Step 1. Probabilities of an event at τ_j: $p_j = S(\tau_{j-1}) - S(\tau_j)$, $j = 1, \ldots, m$.

Step 2. Expected number of events at τ_j: $d_j = \sum_{i=1}^{n} \frac{\alpha_{ij} p_j}{\sum_k \alpha_{ik} p_k}$ (each subject's unit mass is split over its admissible grid points in proportion to p_j).

Step 3. Estimated number at risk: $Y_j = \sum_{k=j}^{m} d_k$.

Step 4. Update the product-limit estimator from these pseudo-data; stop if it is close to the previous S at all τ_j, else repeat Steps 1–3.

Example 8.5: Time to cosmetic deterioration, early breast cancer (radiotherapy only)

46 patients; data are intervals $(L, R]$ in months or right-censored "$\geq$" values: $(0, 7]$; $(0, 8]$; $(0, 5]$; $(4, 11]$; $(5, 12]$; $(5, 11]$; $(6, 10]$; $(7, 16]$; $(7, 14]$; $(11, 15]$; $(11, 18]$; ≥ 15; ≥ 17; $(17, 25]$; $(17, 25]$; ≥ 18; $(19, 35]$; $(18, 26]$; ≥ 22; ≥ 24; ≥ 24; $(25, 37]$; $(26, 40]$; $(27, 34]$; ≥ 32; ≥ 33; ≥ 34; $(36, 44]$; $(36, 48]$; ≥ 36; ≥ 36; $(37, 44]$; ≥ 37; ≥ 37; ≥ 37; ≥ 38; ≥ 40; ≥ 45; ≥ 46 ($\times 7$).

- **Initial estimate**: distribute each subject's mass $1/46$ equally over the grid points in $(L_i, R_i]$; e.g. the subject with $(0, 7]$ gives $(1/46)(1/4)$ to each of $\tau = 4, 5, 6, 7$.
- With S_0: $p_1 = 1 - 0.979 = 0.021$, $p_2 = 0.979 - 0.955 = 0.024$, etc.
- At $\tau = 4$, three intervals contain it ($(0, 8]$, $(0, 7]$, $(0, 5]$), so

$$d_1 = \frac{0.021}{0.021 + 0.024 + 0.021 + 0.029 + 0.031} + \frac{0.021}{0.021 + 0.024 + 0.021 + 0.029} + \frac{0.021}{0.021 + 0.024} = 0.842.$$

- One iteration changes S only slightly at early times (e.g. $0.979 \to 0.982$ at $\tau = 4$) but more in the tail (e.g. $0.229 \to 0.341$ at $\tau = 46$); iterate to convergence.

4 The Cohort Life Table

4.1 Setting

A **cohort** is a group of individuals with a common origin from which event times are measured. Each event or censoring time is recorded only as falling in one of $k + 1$ adjacent, non-overlapping intervals $(a_{j-1}, a_j]$, $j = 1, \ldots, k + 1$. A traditional cohort life table presents the actual mortality experience of the cohort from the birth of each individual to the death of the last surviving member; it summarizes time-to-event information over these intervals (grouped data).

Example 8.6: NLSY weaning data

National Labor Survey of Youth (K&M Section 1.14): 927 first-born infants who were breast fed and have complete information. Response: duration of breast feeding in weeks, with an indicator of whether the baby was weaned (event) or lost to follow-up (censored). First rows: weeks 0–2: 927 entering, 2 withdrawn; 2–3: 848 entering, 3 withdrawn; 3–5: 774 entering, 6 withdrawn; ...

4.2 Construction

Definition 8.7: Life-table quantities

1. Intervals $I_j = (a_{j-1}, a_j]$, $j = 1, \ldots, k+1$, $a_0 = 0$, $a_{k+1} = \infty$; each event/censoring time falls in exactly one of them.
2. Y'_j: number entering I_j without having experienced the event.
3. W_j: number lost to follow-up or withdrawn alive in I_j.
4. $Y_j = Y'_j - W_j/2$: estimated number at risk in I_j, assuming censoring times are uniform over the interval (each withdrawal is at risk for half the interval on average) — the *actuarial* adjustment.
5. d_j: number of events in I_j.
6. Survival at the start of I_j ($\hat{S}(a_0) = 1$):

$$\hat{S}(a_j) = \hat{S}(a_{j-1})[1 - d_j/Y_j] = \prod_{i=1}^{j} [1 - d_i/Y_i]. \tag{8.1}$$

7. Density at the midpoint $a_{mj} = (a_{j-1} + a_j)/2$:

$$\hat{f}(a_{mj}) = \frac{\hat{S}(a_{j-1}) - \hat{S}(a_j)}{a_j - a_{j-1}}. \tag{8.2}$$

8. Hazard at the midpoint (with $\hat{S}(a_{mj})$ taken as the average of the endpoint values):

$$\hat{h}(a_{mj}) = \frac{\hat{f}(a_{mj})}{\hat{S}(a_{mj})} = \frac{2\hat{f}(a_{mj})}{\hat{S}(a_j) + \hat{S}(a_{j-1})} = \frac{d_j}{(a_j - a_{j-1})(Y_j - d_j/2)}. \tag{8.3}$$

Derivation: deriving (8.3)

Write $\hat{q}_j = d_j/Y_j$, $\hat{p}_j = 1 - \hat{q}_j$ and $\Delta_j = a_j - a_{j-1}$. Then $\hat{S}(a_j) = \hat{S}(a_{j-1})\hat{p}_j$ and $\hat{f}(a_{mj}) = \hat{S}(a_{j-1})\hat{q}_j/\Delta_j$, so

$$\hat{h}(a_{mj}) = \frac{2\hat{S}(a_{j-1})\hat{q}_j/\Delta_j}{\hat{S}(a_{j-1})(1 + \hat{p}_j)} = \frac{2\hat{q}_j}{\Delta_j(1 + \hat{p}_j)} = \frac{2d_j/Y_j}{\Delta_j(2 - d_j/Y_j)} = \frac{d_j}{\Delta_j(Y_j - d_j/2)}.$$

Interpretation: events per unit of person-time, where those who fail in I_j are credited with half the interval.

4.3 Standard errors

Theorem 8.8: Estimated standard deviations

$$\widehat{\mathrm{SD}}[\hat{S}(a_{j-1})] = \hat{S}(a_{j-1})\sqrt{\sum_{i=1}^{j-1}\frac{d_i}{Y_i(Y_i-d_i)}} = \hat{S}(a_{j-1})\sqrt{\sum_{i=1}^{j-1}\frac{\hat{q}_i}{Y_i\hat{p}_i}}, \tag{8.4}$$

$$\widehat{\mathrm{SD}}[\hat{f}(a_{mj})] = \frac{\hat{S}(a_{j-1})\hat{q}_j}{a_j-a_{j-1}}\sqrt{\sum_{i=1}^{j-1}\frac{\hat{q}_i}{Y_i\hat{p}_i}+\frac{\hat{p}_j}{Y_j\hat{q}_j}}, \tag{8.5}$$

$$\widehat{\mathrm{SD}}[\hat{h}(a_{mj})] = \left\{\frac{1-[\hat{h}(a_{mj})(a_j-a_{j-1})/2]^2}{Y_j\hat{q}_j}\right\}^{1/2}\hat{h}(a_{mj}). \tag{8.6}$$

(8.4) is Greenwood's formula; the SD of $\hat{S}(a_0)=1$ is 0.

4.4 Other summary quantities

1. **Mean**: $\hat{\mu}=\int_0^\infty \hat{S}(t)\,\mathrm{d}t$. If the largest observation is censored, $\hat{S}$ stays flat and the tail must be completed in one of the three ways of Lecture 5 (Efron, Gill, Brown–Hollander–Kowar).
2. **Median** by linear interpolation: find I_j with $\hat{S}(a_{j-1})\ge 0.5\ge\hat{S}(a_j)$, then
$$\hat{x}_{0.5}=a_{j-1}+\frac{[\hat{S}(a_{j-1})-0.5](a_j-a_{j-1})}{\hat{S}(a_{j-1})-\hat{S}(a_j)}=a_{j-1}+\frac{\hat{S}(a_{j-1})-0.5}{\hat{f}(a_{mj})}. \tag{8.7}$$
3. **Mean residual life** at the start of I_i:
$$\widehat{\mathrm{mr}}(a_{i-1})=\frac{\int_{a_{i-1}}^\infty \hat{S}(t)\,\mathrm{d}t}{\hat{S}(a_{i-1})}=\frac{\sum_{j=i}^k \hat{S}(a_{j-1})(a_j-a_{j-1})}{\hat{S}(a_{i-1})}.$$
4. **Median residual life**: $\mathrm{mdrl}(a_{i-1})$ solves $S(a_{i-1})/2 = S(a_{i-1}+\mathrm{mdrl}(a_{i-1}))$ — the time increment beyond which half of those alive at a_{i-1} are expected to survive.

Example 8.9: life table for weaning (K&M Table 5.6)

Weeks	Y'_j	W_j	Y_j	d_j	$\hat{S}(a_{j-1})$	$\hat{f}(a_{mj})$	$\hat{h}(a_{mj})$	SD $\hat{S}$	SD $\hat{f}$	SD $\hat{h}$
0–2	927	2	926	77	1.0000	0.0416	0.0434	0	0.0045	0.0049
2–3	848	3	846.5	71	0.9168	0.0769	0.0875	0.0091	0.0088	0.0104
3–5	774	6	771	119	0.8399	0.0648	0.0836	0.0121	0.0055	0.0076
5–7	649	9	644.5	75	0.7103	0.0413	0.0618	0.0149	0.0046	0.0071
7–11	565	7	561.5	109	0.6276	0.0305	0.0537	0.0160	0.0027	0.0051
11–17	449	5	446.5	148	0.5058	0.0279	0.0662	0.0166	0.0021	0.0053
17–25	296	3	294.5	107	0.3381	0.0154	0.0555	0.0158	0.0014	0.0052
25–37	186	0	186	74	0.2153	0.0071	0.0414	0.0138	0.0008	0.0047
37–53	112	0	112	85	0.1296	0.0061	0.0764	0.0114	0.0006	0.0066
53–	27	0	27	27	0.0313	—	—	0.0059	—	—

Check of row 1: $Y_1 = 927-2/2 = 926$; $\hat{q}_1 = 77/926 = 0.0832$, so $\hat{S}(2)=0.9168$; $\hat{f}=(1-0.9168)/2=0.0416$; $\hat{h}=77/\{2(926-38.5)\}=0.0434$.

Median: $\hat{S}(11)=0.5058\ge 0.5\ge\hat{S}(17)=0.3381$, so $\hat{x}_{0.5}=11+\dfrac{(0.5058-0.5)(17-11)}{0.5058-0.3381}\approx$ 11.2 weeks.

Hazard (K&M Fig. 5.3): high (about 0.08–0.09 per week) in the first 5 weeks, about 0.05–0.07 through week 25, lowest (about 0.04) in weeks 25–37, then rising again in weeks 37–53.

5 Selected exercises

Motivation. Turnbull's algorithm and the life table were presented as recipes. These exercises show what the algorithm converges to, where the actuarial hazard (8.3) comes from, and how its standard error (8.6) follows from the delta method.

Exercise 8.1 (Turnbull's algorithm and the likelihood)

In the interval-censored setting, let $p = (p_1, \dots, p_m)$ range over the simplex $\Delta = \{p_j \geq 0,\ \sum_j p_j = 1\}$, assume each subject has some $\alpha_{ij} = 1$, and let

$$\ell(p) = \sum_{i=1}^{n} \log\Big(\sum_{j=1}^{m} \alpha_{ij} p_j\Big), \qquad D_j(p) = \frac{\partial \ell}{\partial p_j} = \sum_{i=1}^{n} \frac{\alpha_{ij}}{\sum_k \alpha_{ik} p_k},$$

the log-likelihood (subject i contributes $P(L_i < X \leq R_i)$) and its partial derivatives, at p with $\ell(p) > -\infty$.
(a) Show that one pass of Steps 1–4 maps p to $T(p)$ with $T(p)_j = d_j/n = p_j D_j(p)/n$.
(b) Show that every maximizer of ℓ over Δ is a fixed point of T.
(c) Conversely, show that a fixed point p of T with $D_j(p) \leq n$ for every j with $p_j = 0$ maximizes ℓ over Δ.

Solution [Supplement].

(a)

Step 1. By Step 2 of the algorithm, $d_j = p_j \sum_i \alpha_{ij} / \sum_k \alpha_{ik} p_k = p_j D_j$. Summing over j, each subject contributes $\sum_j \alpha_{ij} p_j / \sum_k \alpha_{ik} p_k = 1$, so $\sum_j d_j = n$; equivalently $\sum_j p_j D_j = n$.

Step 2. The pseudo-data have no censoring: $Y_j = \sum_{k \geq j} d_k$, so $Y_1 = n$ and $Y_{j+1} = Y_j - d_j$ (with $Y_{m+1} = 0$). As in Exercise 5.1(a), the product-limit estimate telescopes to $S_{\text{new}}(\tau_j) = Y_{j+1}/n$ (if some $Y_l = 0$, the previous factor is already 0). Hence $S_{\text{new}}(\tau_{j-1}) - S_{\text{new}}(\tau_j) = (Y_j - Y_{j+1})/n = d_j/n$.

(b)

Step 1. Let p maximize ℓ; then $\ell(p) > -\infty$, so every $\sum_k \alpha_{ik} p_k > 0$. For $q \in \Delta$ and $\varepsilon \in [0, 1)$ these sums stay positive along $p + \varepsilon(q - p)$, and maximality gives the one-sided directional derivative

$$\sum_j D_j(p)(q_j - p_j) = \sum_j D_j q_j - n \leq 0.$$

Step 2. With $q = e_j$ (the j-th vertex): $D_j \leq n$ for all j.

Step 3. If $0 < p_j < 1$, take $q = (p - p_j e_j)/(1 - p_j) \in \Delta$. Then $\sum_k D_k q_k = (n - p_j D_j)/(1 - p_j)$, and Step 1 gives $p_j(n - D_j)/(1 - p_j) \leq 0$, i.e. $D_j \geq n$. If $p_j = 1$, then $D_j = n$ by $\sum_k p_k D_k = n$. So $D_j = n$ whenever $p_j > 0$.

Step 4. By (a), $T(p)_j = p_j D_j / n = p_j$ if $p_j > 0$, and $T(p)_j = 0 = p_j$ if $p_j = 0$.

(c)

Step 1. $T(p) = p$ means $p_j D_j = n p_j$, so $D_j = n$ where $p_j > 0$; with the hypothesis, $D_j \leq n$ for all j.

Step 2. Let $q \in \Delta$ with $\ell(q) > -\infty$ (otherwise there is nothing to prove). The function $g(\varepsilon) =$

$\ell(p+\varepsilon(q-p))$ is concave on $[0,1]$ (a sum of logs of positive affine functions), so $g(1) \le g(0)+g'(0+)$:

$$\ell(q) \le \ell(p) + \sum_j D_j(q_j - p_j) \le \ell(p) + n\sum_j q_j - n = \ell(p).$$

So Turnbull's iteration is a fixed-point (EM) scheme for the nonparametric MLE; a limit with some $p_j = 0$ must still be checked against $D_j \le n$. ■

Exercise 8.2 (The actuarial hazard is a piecewise-exponential MLE)

Suppose the hazard is constant on each life-table interval, $h(x) = h_j$ for $x \in I_j$, and right-censored data (T_i, δ_i) are observed exactly.
(a) Show that the MLE is $\hat h_j = d_j/E_j$, where $E_j = \sum_i e_{ij}$ is the total time at risk in I_j, $e_{ij} = \max\{0, \min(T_i, a_j) - a_{j-1}\}$.
(b) If every death and withdrawal in I_j is placed at the midpoint of I_j, show that $\hat h_j$ equals (8.3).

Solution [Supplement].

(a)

Step 1. By the right-censoring likelihood of Lecture 4, subject i contributes $h(T_i)^{\delta_i} e^{-H(T_i)}$, and $H(T_i) = \int_0^{T_i} h = \sum_j h_j e_{ij}$. If $\delta_i = 1$ and $T_i \in I_j$, then $h(T_i) = h_j$.

Step 2. Multiplying over i,

$$L(h_1, \dots, h_{k+1}) = \prod_j h_j^{d_j} e^{-h_j E_j}, \qquad \log L = \sum_j \{d_j \log h_j - h_j E_j\}.$$

Each term is strictly concave in h_j when $d_j > 0$, with derivative $d_j/h_j - E_j$ vanishing at $h_j = d_j/E_j$; when $d_j = 0$ the term $-h_j E_j$ is maximized at $h_j = 0 = d_j/E_j$.

(b) Of the Y_j' subjects entering I_j, $Y_j' - d_j - W_j$ survive the interval (exposure Δ_j each), and the $d_j + W_j$ who die or withdraw contribute $\Delta_j/2$ each. Thus

$$E_j = \Delta_j\Big\{Y_j' - d_j - W_j + \frac{d_j + W_j}{2}\Big\} = \Delta_j\Big(Y_j' - \frac{W_j}{2} - \frac{d_j}{2}\Big) = \Delta_j\Big(Y_j - \frac{d_j}{2}\Big),$$

and $\hat h_j = d_j/\{\Delta_j(Y_j - d_j/2)\}$, which is (8.3). ■

Exercise 8.3 (Standard error of the life-table hazard)

Treat $\hat q_j = d_j/Y_j$ as a binomial proportion, $\mathrm{Var}(\hat q_j) \approx \hat q_j \hat p_j / Y_j$. Derive (8.6) by the delta method.

Solution [Supplement].

Step 1. By the derivation of (8.3), $\hat h = g(\hat q)$ with $g(q) = \dfrac{2q}{\Delta(2-q)}$ (index j dropped), and

$$g'(q) = \frac{2(2-q)+2q}{\Delta(2-q)^2} = \frac{4}{\Delta(2-q)^2}, \qquad \widehat{\mathrm{SD}}(\hat h) \approx \frac{4}{\Delta(2-\hat q)^2}\Big(\frac{\hat q \hat p}{Y}\Big)^{1/2}.$$

Step 2. Put $u = \hat h \Delta/2 = \hat q/(2-\hat q)$. Then $1 + u = 2/(2-\hat q)$ and $1 - u = 2\hat p/(2-\hat q)$, so $1 - u^2 = 4\hat p/(2-\hat q)^2$.

Step 3. The right-hand side of (8.6) is therefore

$$\hat h\Big(\frac{1-u^2}{Y\hat q}\Big)^{1/2} = \frac{2\hat q}{\Delta(2-\hat q)} \cdot \frac{2}{2-\hat q}\Big(\frac{\hat p}{Y\hat q}\Big)^{1/2} = \frac{4}{\Delta(2-\hat q)^2}\Big(\frac{\hat q \hat p}{Y}\Big)^{1/2},$$

the expression of Step 1. ■